Data Fundamentals & Data Management

Session Objective

Understand:

- Different types of data
- Data sources
- Data collection methods
- Data lifecycle
- Data quality and integrity

Also understand how companies use data in real-time business operations.

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1. Understanding Different Data Types (25 Minutes)

Data means information collected by companies for business purposes.

Example – Online Shopping

When you order a product online, the company collects:

- Customer name
- Address
- Product details
- Payment information

- Delivery status

This data helps the company:

- Deliver products
- Track orders
- Analyze sales
- Improve customer service

Simple Conclusion

Data is important because companies use it to:

- Make business decisions
- Improve services
- Increase sales
- Understand customers
- Generate reports and dashboards

Without data, companies cannot properly manage their operations.

A. Structured Data

Structured data is organized in rows and columns.

Examples

- Excel sheets
- SQL database tables
- Employee records

Emp ID	Name	Salary
101	Ravi	25000

Features

- Easy to search
- Easy to analyze
- Stored in databases

Real-Time Example

Bank customer account details.

B. Unstructured Data

Unstructured data has no fixed format.

Examples

- Images
- Videos
- Audio files
- Social media posts

Real-Time Example

Hospital X-ray images or CCTV recordings.

C. Semi-Structured Data

Semi-structured data is partially organized.

Examples

- JSON files
- XML files
- Emails

Real-Time Example

Data received from APIs.

Example:

```None
{

 "name":"Ravi",

 "age":25

}
```

XML Example (eXtensible Markup Language)

XML stores data using tags.

Example

```None

<Employee>

  <EmpID>101</EmpID>

  <Name>Ravi</Name>

  <Department>HR</Department>

</Employee>
```

Explanation

- `<EmpID>` stores employee ID
- `<Name>` stores employee name
- `<Department>` stores department

Here data is organized using tags, so it is called semi-structured data.

2. Common Data Sources (25 Minutes)

Data can come from different places.

A. Internal Data Sources

Data generated inside the company.

Examples

- Employee records
- Sales reports
- Inventory data

Real-Time Example

Hospital patient database.

B. External Data Sources

Data collected from outside the company.

Examples

- Social media
- Government websites
- Market reports

Real-Time Example

Weather data used by delivery companies.

C. Databases

Database stores data in structured format.

Popular Databases

- MySQL
- Oracle Database

Example

Customer details stored in banking systems.

D. APIs (Application Programming Interfaces)

API is a method that allows two applications or systems to communicate and share data with each other.

Simple Meaning

API acts like a messenger between applications.

One application sends a request, and another application sends a response.

Simple Real-Time Example – Food Delivery App

When you order food in an app:

1. App sends order request to restaurant system using API
2. Payment gateway confirms payment using API
3. Google Maps API tracks delivery location
4. Customer receives live order updates

All these systems communicate through APIs.

Easy Analogy

Restaurant Example

Imagine:

- You are the customer
- Kitchen is the server/system
- Waiter is the API

You tell the waiter your order.
The waiter takes it to the kitchen and brings back the food.

Similarly:

- User sends request
- API transfers request
- Server sends response

How API Works

Step-by-Step Flow

User/Application → API → Server/Database → API → Response to User

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Real-Time API Examples

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1. Payment API

Used in:

- PhonePe
- Google Pay

What It Does

- Processes payments
- Verifies bank details
- Confirms transaction status

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2. Google Maps API

Used in:

- Delivery apps
- Cab booking apps

Functions

- Shows live location
- Finds routes
- Calculates distance

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3. Weather API

Weather apps collect:

- Temperature
- Rain information
- Climate updates

from weather servers using APIs.

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4. Login API

Many apps allow:

- Login with Google
- Login with Facebook

This happens using APIs.

API in Business Systems

ERP + CRM + API Connection

Example – Online Shopping Company

CRM

Stores customer details.

ERP

Handles inventory and billing.

API

Connects:

- Payment system
- Delivery system
- SMS notifications

All systems work together using APIs.

Types of APIs (Basic)

Type	Purpose
Payment API	Online payments
Maps API	Location tracking
SMS API	Sending messages
Weather API	Live weather data
Authentication API	Login verification

API Data Format

APIs usually send data in:

- JSON
- XML

JSON Example

```json
{
 "name":"Ravi",
 "city":"Bangalore"
}
```

Benefits of APIs

- Faster communication
- Automation
- Real-time data sharing
- Connects multiple systems
- Reduces manual work

3. Data Collection Methods (20 Minutes)

Data collection means gathering information from different sources.

Common Methods

1. Forms

Example:

- Online registration forms
- Hospital patient forms

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2. Surveys

Example:

- Customer feedback surveys

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3. Sensors & Devices

Example:

- Smart watches collecting health data
- ATM machines recording transactions

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4. Websites & Applications

Example:

- Online shopping websites collect customer activity

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5. APIs

Example:
Apps collect live weather or payment data through APIs.

4. Data Lifecycle Management (30 Minutes)

Data lifecycle means how data moves from creation to deletion.

Stages of Data Lifecycle

1. Data Creation

Data is created when:

- Customer places an order
- Employee joins company
- Payment happens

Example

Patient registration in hospital.

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2. Data Storage

Data is stored in:

- Databases
- Cloud systems
- Servers

Example

Bank stores account information securely.

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3. Data Usage

Data is used for:

- Reports
- Analysis
- Decision making

Example

Sales reports created using Power BI.

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4. Data Archiving

Old data is moved to archive storage.

Example

Old hospital records stored separately.

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5. Data Deletion

Unused or expired data is deleted securely.

Example

Deleting inactive customer accounts.

Real-Time Lifecycle Example – E-commerce Company

1. Customer places order → Data created
2. Order stored in database → Data storage
3. Sales dashboard created → Data usage
4. Old orders archived → Archiving
5. Unnecessary data removed → Deletion

5. Data Quality and Integrity (20 Minutes)

Good quality data is very important for businesses.

Data Quality

Data should be:

- Accurate
- Complete
- Consistent
- Reliable

Bad Data Example

- Wrong phone numbers

- Duplicate records
- Missing values

Data Integrity

Data integrity means data should remain correct and unchanged.

Example

Bank transaction amount should not change incorrectly.

Real-Time Problems Due to Bad Data

Example – Hospital

Wrong patient data may:

- Generate wrong reports
- Cause billing issues
- Affect treatment

How Companies Maintain Data Quality

- Data validation
- Removing duplicates
- Checking null values
- Data cleaning using Excel/SQL

Technologies Used in Data Management

Tool	Purpose
Excel	Data cleaning and reports
SQL	Database handling
Power BI	Dashboards and analysis
APIs	Data transfer
ERP	Business operations data
CRM	Customer data

Career Opportunities Related to These Topics

Job Role	Skills Required
Data Analyst	Excel, SQL, Power BI
Data Processing Executive	Data cleaning
Business Analyst	Business understanding
SQL Developer	Database handling
Reporting Analyst	Dashboard creation